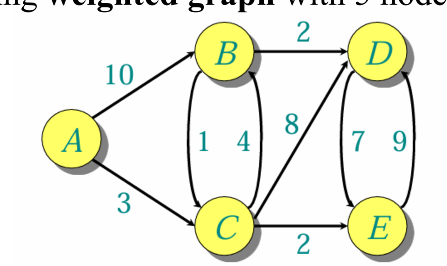
<u>Assignment - 1</u>

Subject: DAA
Deadline: 1 September 2025

Q1. Define algorithm? What are its characteristics?
Q2. Write the algorithm of Merge sort and find its time complexity.
Q3. Explain asymptotic notation (Big Oh, Big theta, Big Omega).
Q4. Find out the time complexity of binary Search.
Q5.Explain Prim's and Kruskal's algorithm with the help of suitable examples.
Q6. Sort the following number using Quick sort : 4,3,7,1,8,5,9,6
Q7.You have a knapsack with a **capacity of 50 units**. The following items are available:

Item	Profit	Weight
1	70	31
2	100	33
3	120	45
4	80	25
5	30	15

Find the **maximum profit** using the fractional knapsack approach. Show which fractions of items are selected.
Q8. Given the following **weighted graph** with 5 nodes: **A, B, C, D, E**



Find- (i). Distance table after each iteration.
(ii).Final shortest path from A to D
(iii) Total minimum cost
Q9. What is the time complexity of the dynamic programming solution for matrix chain multiplication? Explain why.
Q10.Explain the time complexity of the naive recursive Fibonacci algorithm. Why is it so inefficient?